

Flávio Silva

Braga, Portugal
LinkedIn | Github

ABOUT ME

Second year Master's Degree student in Software Engineering at the University of Minho in Braga, Portugal, with hands-on experience in full-stack development, systems programming, and containerized deployment.

SKILLS

C, Haskell, JavaScript, TypeScript, Python, Vue.js, SQL, Java, Next.js, React, Node.js, Git, Docker, REST APIs, Full Stack Development

PROJECTS

F3M_APP | Node.js / React / SQL

2025

- Full-stack **mobile application** developed for the **Informatics Project** course at UMinho, in collaboration with company **F3M**, designed to assist patients with mild to moderate **dementia** in **medication management**, in the **health tech / e-health** domain.
- Built **RESTful API** endpoints and a **relational SQL database** to allow caregivers to manage medication schedules, prescriptions, and pill reminders through a dedicated **alarm/notification system**.
- Developed an information section providing caregivers with guidance on **emergency procedures** for various dementia-related scenarios, contributing to **user experience (UX)** design.
- [Github] (https://github.com/MartimRedondo/F3M_APP)

PictuRas | Vue.js

2025

- A scalable **web application** designed to manage, edit, and showcase images effectively, built with a modern **frontend framework** and **RESTful API** backend.
- Includes a **Vue.js** frontend for user interaction, a backend API service, and integrated **email management/testing** with **Mailhog**.
- Users can apply various **image processing** tools and filters directly within the application. The project is deployed using **Docker** containers and **Kubernetes**, leveraging **Minikube** and **Helm** for container orchestration and package management, following **DevOps** and **CI/CD** practices.
- [Github] (<https://github.com/JoaoCoelho2003/PictuRas>)

Server Client System | C

2023

- Designed and implemented a **Server/Client architecture** using **Pipes** and **Forks** for **inter-process communication (IPC)** and **concurrent programming**.
- The Server is designed to handle **multiple concurrent Client requests**, including executing one or more Programs Concurrently or performing a status check on Scheduled or Completed Programs, along with **execution time metrics**.
- This Server also implements a custom “mysystem” command integrated into the C Library, built using **forks and execs** for **process management** on **Unix/Linux** systems.
- [Github] (<https://github.com/FuzzyLaDuzzy/SOTP-2024>)

EDUCATION

University of Minho, Portugal

2020-2024

Bachelor's Degree in Software Engineering

University of Minho, Portugal

2024-Present

Master's Degree in Software Engineering